

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
MID TERM EXAMINATION SEPTEMBER 2024



Course: B.Tech
 Subject Name: EMEC-I

Semester: III
 Subject Code: EE212
 Date: 04-10-2024
 Time: 90 mins.

Maximum Marks: 20
 Course Coordinators: Dr. Priyanka Paliwal and Dr. Ch Hemanth

Instructions: Answer all the following questions

Q. No.	Questions	Marks	COs
1.	Two watt meters are used to measure power and power factor of a 3-phase delta connected load having a 25Ω resistor in series with a 15mH inductor in each phase. The load is supplied from a 415 V , 50Hz mains. Determine the readings of the watt meters.	04	CO 1
2.	A 220 V , 3-phase voltage is applied to a balanced delta connected load of phase impedance $(6+j8)$. (i) Find the phasor current in each line (ii) What is the power consumed per phase? (iii) What is the phasor sum of three line currents? Why does it have this value?	04	CO 1
3.	Two similar 250 kVA , single-phase transformers gave the following result when tested by Sumpner's test: Wattmeter W_1 (Connected to mains) = 5 kW W_2 (at full load current) = 7.5 kW . Find out the individual transformer efficiencies at 75% full load and 0.8 pf lag.	04	CO 2
4.	A 20 kVA , $2400/240 \text{ V}$ two winding transformer is to be used as an auto transformer. Calculate the kVA rating of auto transformer, transformed kVA and conducted kVA for both series additive polarity and series subtractive polarity?	04	CO 2
5. a.	An industrial load takes 100 A at 0.8 p.f. lag from a 3-phase $11000/400 \text{ V}$, 50 Hz $\text{Y}-\Delta$ transformer. Calculate the kVA rating of transformer, phase currents on both l.v. side and h.v. sides?	03	CO 2
b.	A $220/132 \text{ kV}$ Y/Δ transformer is reconnected to Y/Y , what is new voltage rating of transformer?	01	CO 2